



Avoiding Racial Bias in Child Welfare Agencies' Use of Predictive Risk Modeling

Actions child welfare agencies can take to mitigate the likelihood that predictive risk models will exacerbate racial and ethnic disparities include carefully examining the data used to develop these models, evaluating model performance and impact, engaging community members throughout the process, and achieving transparency about how the models are used.

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KEY POINTS

- Several child welfare agencies in the United States have implemented or are developing predictive risk models (PRMs) to help inform child welfare decisions.
- PRMs can assist caseworkers by identifying previously unobserved patterns in data related to future outcomes. However, PRMs also have the potential to exacerbate existing racial and ethnic disparities if not developed with careful attention to the methods used.
- Child welfare agencies and their vendors can take various actions to identify and mitigate racial and ethnic bias when planning, developing, implementing, and monitoring a PRM.
- When planning to develop a PRM, agencies should identify an appropriate use case and available data sources to use in supporting the PRM development.
- When developing a PRM, agencies and vendors should assess the quality of the available data sources, make careful decisions about PRM parameters, validate the outcome variable they plan to predict, and assess its predictive performance for important subgroups.
- When implementing a PRM, agencies should assess the risk and potential benefits of using the model, consider how best to balance PRM misclassification rates, and determine how to present predictions to agency staff and train them in using those predictions.
- When monitoring a PRM that has been implemented, agencies should continuously examine its predictive performance and evaluate its impact on equity and other outcomes.
- In each phase, agencies should engage community members and caseworkers in making key decisions and be transparent about their approach to developing and using PRMs.
- Federal agencies could consider supporting the integration of equitable, high-quality PRMs by (1) refusing to fund the development of proprietary PRMs; (2) developing guidelines on approaches for engaging with community members, governing the use of PRMs, and the public release of information about how PRMs are being used; and (3) working with child welfare agencies to conduct rigorous evaluations of these models.

INTRODUCTION

Overview of Child Welfare in the United States

In 2020, about 3.9 million referrals involving about seven million children were made to child protective services (CPS) across the United States (U.S. Department of Health and Human Services, Administration for Children and Families, Administration on Children, Youth and Families, Children's Bureau, 2022a). That same year, there were just over 400,000 children in foster care (U.S. Department of Health and Human Services, Administration for Children and Families, Administration on Children, Youth and Families, Children's Bureau, 2022b). Overall, about 37% of all children in the United States experience a CPS investigation by the age of 18. Rates of child welfare investigations are greatest for certain racial and ethnic groups, with African American and Hispanic children experiencing an investigation by the age of 18 at rates of approximately 53% and 32%, respectively, compared to 28% of White children (Kim et al., 2017). Additionally, children from certain racial and ethnic groups are overrepresented in the foster care system. For instance, while American Indian and Alaska Native children and African American children made up just 1% and 14% of the U.S. child population in 2019, respectively, they accounted for 2% and 23% of the foster care population (Child Welfare Information Gateway, 2021).

Child welfare agencies receiving referrals for child maltreatment must make a number of decisions about them, as well as the families already involved in child welfare services. These decisions include the following: (1) screening decisions based on referral calls, including deciding whether an investigation is warranted and triaging investigations based on child safety or maltreatment risk classification; (2) decisions about removing children from their families based on the results of those investigations; (3) decisions about moving children from one foster care placement to another; (4) decisions about returning children to their families; (5) assessments of risk of needing foster care or child welfare services; and (6) decisions about allocating resources and interventions across communities (Vaithianathan, Putnam-Hornstein et al., 2019). The decisions that child welfare agencies make can have a significant impact on children, families, and communities; in addition, child welfare decision-making processes can vary substantially across states and regions (Casey Family Programs, 2018; Damman et al., 2020).

Although evidence shows that families from certain racial and ethnic groups are overrepresented in their involvement with CPS (U.S. Department of Health and Human Services, Administration for Children and Families, Administration on Children, Youth and Families, Children's Bureau, 2022a), there are outstanding questions related to the extent to which this pattern is the result of biased decision-making on the part of child welfare agencies. Researchers have developed six main explanatory pathways for this disproportionality: (1) disproportionate and disparate needs of children of different racial and ethnic backgrounds; (2) racial bias and discrimination by individuals, such as protective services caseworkers and mandated reporters; (3) child welfare system factors, such as lack of resources for families of different racial and ethnic backgrounds; (4) geographic context; (5) policy and legislation; and (6) structural racism (Child Welfare Information Gateway, 2021). However, there is a lack of evidence demonstrating the influence various social factors have had on this overrepresentation compared to the influence of the decision-making processes of individual child welfare agencies.

Evolution of Decision-Making Processes in Child Welfare Settings

Child welfare agencies have traditionally relied heavily on human decision making alone to make determinations about the risk of child maltreatment and adverse outcomes (Cuccaro-Alamin et al., 2017). However, given the large number of referrals and limited resources, these agencies can experience challenges identifying and protecting children at risk (Vaithianathan, Putnam-Hornstein et al., 2019). As a result, over the last three decades, agencies have adopted various tools designed to assist staff in making informed and efficient decisions that ultimately reduce the risks of negative outcomes for children and families.

In the early 1990s, child welfare agencies began using consensus-based and actuarial tools to inform decisions (Baird et al., 1999). Consensus-based tools rely on expert theories on the factors that lead to child maltreatment (Cuccaro-Alamin et al., 2017; Drake et al., 2020). These tools then were superseded by actuarial tools, which empirically identify, weight, and combine risk factors related to abuse and neglect to classify a family's or individual's risk of these outcomes (Shlonsky and Wagner, 2005). Research on the use of actuarial tools has shown that those child welfare agencies using them achieve "greater predictive validity and interrater reliability" than agencies not aided by these tools, leading to their wide adoption over the last 20 years (Cuccaro-Alamin et al., 2017; Drake et al., 2020). A limitation of actuarial tools is that variables included in them are those that research has empirically demonstrated to have a relationship to child maltreatment. Therefore, actuarial tools could be missing factors that are predictive of child maltreatment and abuse but do not have a demonstrated relationship to these outcomes (Cuccaro-Alamin et al., 2017; Drake et al., 2020).

More recently, a few researchers and child welfare agencies have begun developing predictive risk models (PRMs) to support child welfare decisions. Lanier et al. (2020) define predictive analytics as a sophisticated form of risk modeling that uses historical data to understand relationships between myriad factors to estimate a probability score for the behavior or outcome of interest. This form of analytics uses machine learning or other data processing techniques to provide information for "decisions, judgments, and/or policy implementations that impact opportunities, access, liberties, rights, and/or safety" (Pittsburgh Task Force on Public Algorithms, 2022). PRMs are intended to assist caseworkers in synthesizing data from various sources to inform decision making (Cheng et al., 2022). However, given the overrepresentation of African American, Hispanic, and American Indian and Alaska Native children in the child welfare system, some have raised concerns that such models may inadvertently incorporate racial and ethnic biases, which could ultimately further increase existing disparities and degrade trust in CPS.

Although PRMs have a variety of potential applications in child welfare, several of the relatively small number of tools currently in use or under development are designed to inform front-end decisions, such as those about the need to investigate referrals made to child welfare hotlines (Exhibit A). This design choice likely has been made because at this initial stage, before any interaction with families occurs, case workers have limited clinical information to inform their decisions but there is a great deal of structured data that a PRM can incorporate to predict what might happen if a particular referral is not investigated. PRMs have also been or are being developed to inform policy decisions related to allocating resources effectively. For example, the Administration for Children's Services in New York City created a PRM to identify risk factors for families who had "frequent involvement" with CPS. It was used for macro-level planning for the child welfare agency's resource allocation (Chapin Hall and Chadwick Center, 2018).